

WASP-49b

R-1309

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_151218 · created 2026-07-31T18:15:50+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457374.8176 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1335 +/- 0.0092
Transit depth (Rp/Rs)^2	1.78 +/- 0.25 [%]
Orbital Inclination (inc)	86.25 +/- 0.72
Airmass coefficient 1 (a1)	1.2715 +/- 0.008
Airmass coefficient 2 (a2)	-0.14 +/- 0.014
Scatter in the residuals of the lightcurve fit is	0.63 %
Best Comparison Star	None
Optimal Aperture	4.52
Optimal Annulus	7.41
Transit Duration (day)	0.1087 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1335 $\pm$ 0.0092 vs 0.116 (deviation 1.9 σ)
Duration fit vs published	0.1087 d vs 0.09 d (deviation 3.46 σ)
Mid-time O-C	3.3 min
Depth SNR	14.5
Residual scatter	0.63 %

Light curve

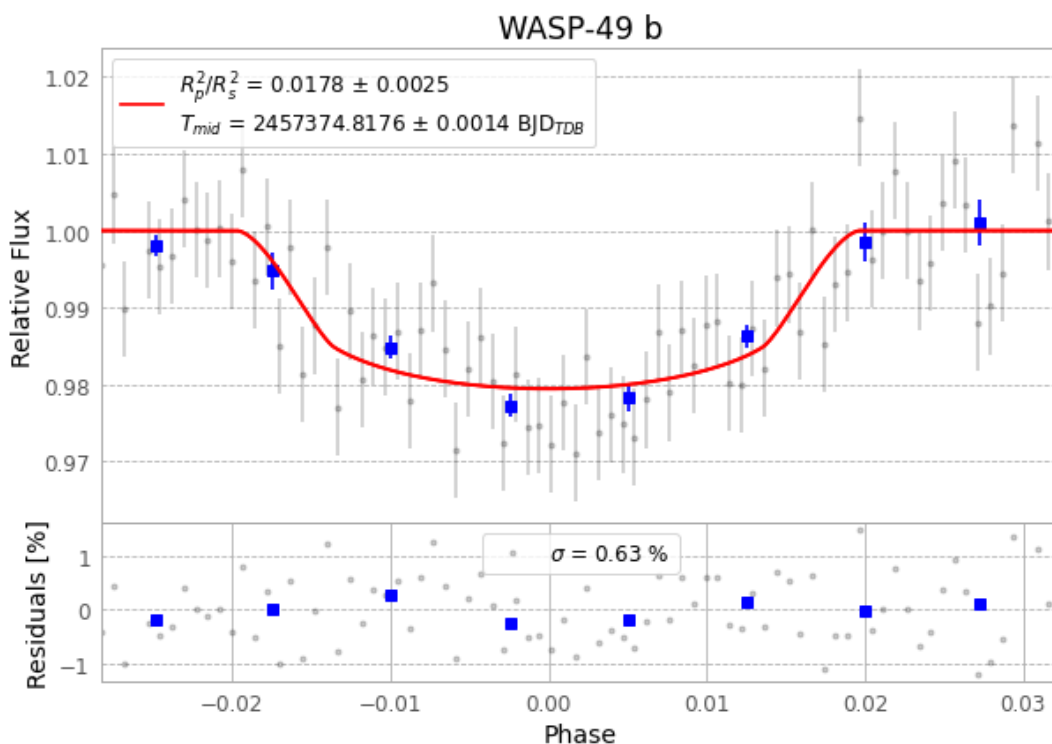


Figure 1 — FinalLightCurve_WASP-49 b_2015-12-17.png (SHA-256 1e657c8157a88628...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [62, 261], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4101eded5d32edc0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), WASP-49151218053612.FITS → WASP-49151218093912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-151218.log (33fd945acbe8...), FinalLightCurve_WASP-49 b_2015-12-17.png (1e657c8157a8...), FinalLightCurve_WASP-49 b_2015-12-17.csv (8779483044cd...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 18:15Z.